# Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation

For ML researchers, AI security practitioners, and cloud infrastructure engineers

Commercial LLM APIs are black boxes: customers cannot directly verify which model is actually serving responses.

This creates an integrity gap between paid model contracts and observable API behavior.

We evaluate whether software-only audits can reliably detect covert model substitution.

Result: they cannot—especially under quantization, nondeterminism, and adversarial routing.

Actionable endpoint: Trusted Execution Environments (TEEs) provide verifiable inference integrity.

# Economic Incentives for Covert Substitution

Why rational providers may silently swap served models

## Cost Pressure

GPU inference dominates OPEX; serving lower-cost models directly increas

## Information Asymmetry

Users observe outputs, not deployed weights, precision, or serving stack det

## Low Detection Risk

Small quality shifts can remain within benchmark variance and user tolera

## High Strategic Payoff

Substitution can preserve pricing while reducing compute expenditure.

# Hypothesis Testing Formulation and Substitution Taxonomy

Can an auditor distinguish $P\_spec(y|x)$ from $P\_actual(y|x)$?

Formal objective: detect whether API outputs come from the contracted model distribution $P\_spec(y|x)$.

Audit challenge: only black-box query access; no direct access to weights, logits pipeline, or hardware state.

Substitution classes: smaller model, precision-reduced variant (e.g., FP8), updated fine-tuned model, or different family.

Practical adversary: a provider can optimize for cost while keeping output drift below detection thresholds.

$$H0: P\_API = P\_spec \quad \text{vs.} \quad H1: P\_API \neq P\_spec$$

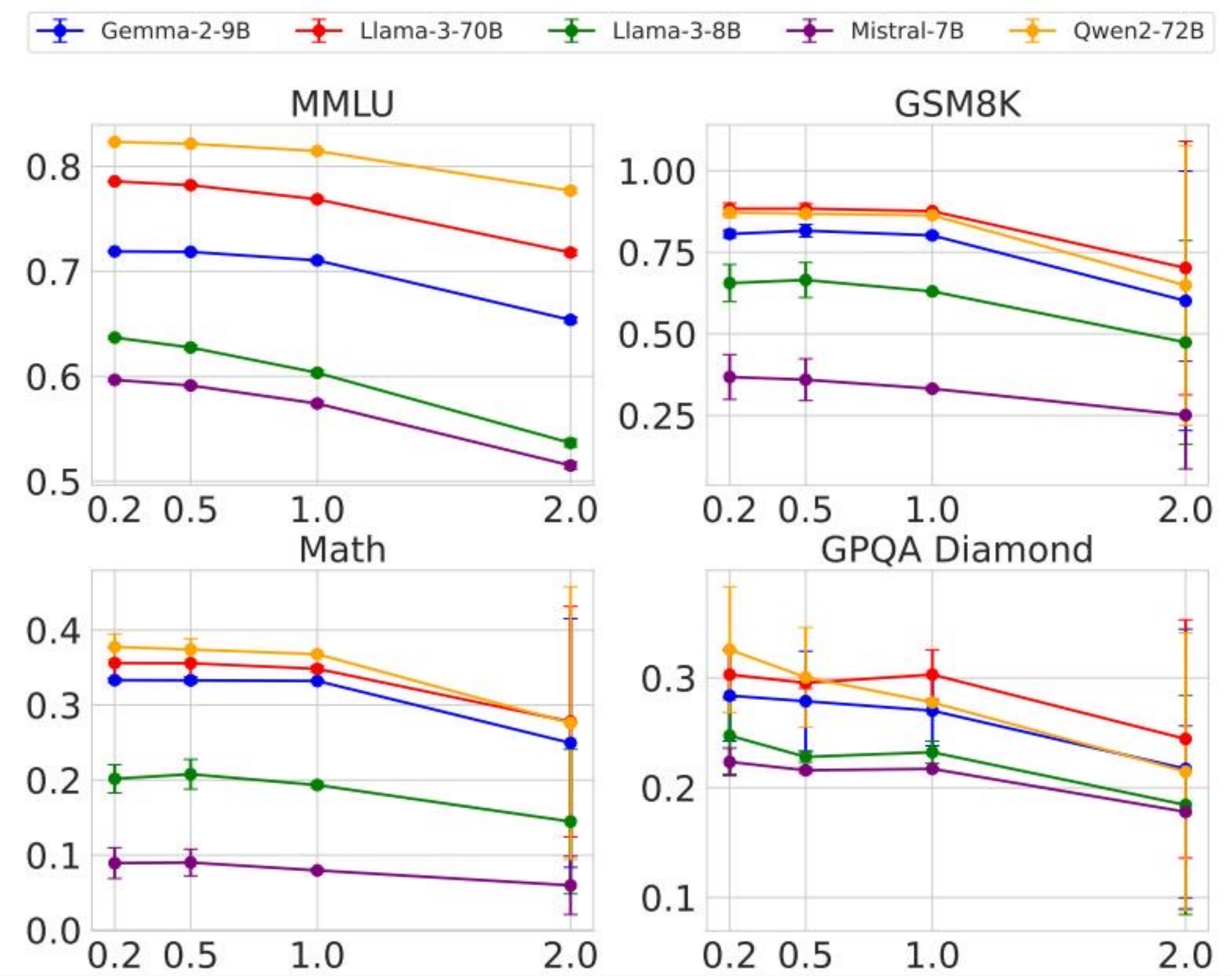
# Quantization Benchmark Drop

Original vs FP8 scores show minimal separability for auditing

| Model Family | Precision | MMLU | GSM8K | MATH | GPQA Diamond |
|---|---|---|---|---|---|
| Llama-3-8B-Instruct | Original | 62.69 ± 0.18 | 61.14 ± 3.47 | 20.65 ± 3.43 | 22.62 ± 0.22 |
| Llama-3-8B-Instruct | FP8 | 62.43 ± 0.26 | 60.90 ± 4.10 | 14.91 ± 2.49 | 20.14 ± 0.24 |
| Llama-3-70B-Instruct | Original | 78.05 ± 0.08 | 88.06 ± 1.44 | 35.69 ± 1.33 | 29.60 ± 0.51 |
| Llama-3-70B-Instruct | FP8 | 77.88 ± 0.13 | 87.35 ± 1.24 | 35.75 ± 1.16 | 33.12 ± 0.30 |
| Gemma-2-9b-it | Original | 71.86 ± 0.08 | 81.80 ± 1.35 | 33.41 ± 0.28 | 28.64 ± 2.97 |
| Gemma-2-9b-it | FP8 | 71.92 ± 0.11 | 79.41 ± 1.14 | 32.53 ± 0.34 | 27.81 ± 3.22 |
| Qwen2-72B-Instruct | Original | 82.18 ± 0.08 | 86.72 ± 1.00 | 37.39 ± 1.41 | 29.93 ± 2.71 |
| Qwen2-72B-Instruct | FP8 | 81.98 ± 0.08 | 86.82 ± 0.97 | 37.67 ± 1.39 | 31.08 ± 1.96 |
| Mistral-7B-Instruct-v0.3 | Original | 59.15 ± 0.10 | 35.90 ± 4.54 | 8.94 ± 1.24 | 21.60 ± 0.17 |
| Mistral-7B-Instruct-v0.3 | FP8 | 58.77 ± 0.13 | 32.20 ± 4.02 | 7.68 ± 1.23 | 22.72 ± 0.19 |

# Temperature Scaling Further Masks Substitution Signals

Reference: figure_1.jpg




Across temperatures (0.2, 0.5, 1.0, 2.0), benchmark accuracy drops broadly for all models.

Rising sampling temperature introduces variance that can dominate subtle model-to-model differences.

Auditors observing only text outputs cannot reliably attribute degradation to substitution versus decoding regime.

Implication: temperature is a confounder that weakens software-only identification power.

# Software-Only Auditing Methods: Failure Matrix

Each method breaks under realistic deployment conditions

| Method Type | Core Weakness | Practical Limitation |
| --- | --- | --- |
| Text-based statistical tests (e.g., MMD) | Distribution gaps are tiny under quantization or mixed routing | Requires very large query budgets; power collapses for subtle substitution |
| Log-probability fingerprinting | Inference-stack and hardware nondeterminism shift token log-probs | Cannot separate legitimate infra variation from malicious model swap |
| Adversarial countermeasures | Provider can randomize substitution or detect benchmark prompts | Benchmark evasion and probabilistic routing make audits query-prohibitive |

# Log-Probability Nondeterminism in Production Inference

Reference: figure_5.jpg

| Model | BERT Acc | T5 Acc | GPT2 Acc | LLM2Vec Acc |
|---|---|---|---|---|
| Llama3-70B-Instruct-FP8 | 50.55 | 50.10 | 50.45 | 49.90 |
| Llama3-70B-Instruct-INT8 | 51.60 | 49.90 | 51.30 | 50.25 |
| Gemma2-9b-it-FP8 | 49.95 | 50.30 | 51.20 | 49.80 |
| Gemma2-9b-it-INT8 | 49.00 | 49.70 | 51.65 | 49.55 |
| Mistral-7b-v3-Instruct-FP8 | 50.55 | 49.75 | 48.70 | 48.75 |
| Mistral-7b-v3-Instruct-INT8 | 49.70 | 50.75 | 50.50 | 51.15 |
| Qwen2-72B-Instruct-FP8 | 50.05 | 50.25 | 48.75 | 49.80 |
| Qwen2-72B-Instruct-INT8 | 50.75 | 50.55 | 49.45 | 50.20 |

Same base model (Llama-3-8B) yields different log-prob outputs across transformers/vLLM versions.

Hardware choice (A100 vs H100) introduces additional measurable variation.

These shifts occur without model substitution, creating false positives and false negatives.

Therefore, log-prob matching is not a stable trust primitive for API integrity.

# Hardware Attestation with TEEs: Practical Verifiable Inference

Provable integrity with deployable overheads

Trusted Execution Environments (TEEs) provide cryptographic attestation of loaded code and model artifacts.

Provider can prove the exact inference binary and model weights running inside confidential hardware boundaries.

Unlike software audits, attestation does not depend on output statistics, prompt sets, or query scale.

Compared with large-model ZKPs, TEEs are operationally feasible today (e.g., NVIDIA confidential computing on H100).

Security posture: provable, efficient, and robust against benchmark gaming and randomized substitution.

# Key Takeaways: Close the Verification Gap with Hardware Attestation

Model substitution is economically rational and difficult to detect from black-box outputs alone.

FP8 quantization often preserves benchmark performance within narrow margins, limiting detection sensitivity.

Inference nondeterminism across frameworks and GPUs invalidates stable log-probability fingerprinting.

Randomized substitution and benchmark-aware routing defeat text-based statistical audits in practice.

Community call to action: treat TEE-backed attestation as baseline infrastructure for trustworthy LLM APIs.

**Auditing by observation is brittle; attestation by hardware is enforceable.**